Lesson plan · for the teacher

# Current, LEDs and resistors

IoT · Level 2 · Lesson 1

**Time**
40 min

**Materials**
Two AA batteries per group, in a holder if you have one, and tape; A torch bulb, or a red LED with a 100 Ω resistor; Three short wires, or strips of kitchen foil; Ten everyday objects: coins, keys, a pencil, a ruler, an eraser, foil

## Learning goals

- Explain that current flows only round a closed loop, pushed by the battery's voltage.
- Light an LED the right way round, with a resistor to limit the current.
- Use Ohm's law to choose a resistor, and spot a short circuit.

## Activity: Conductor hunt

1. Warm-up (5 min): Show a torch with a loose wire or a battery in backwards. Ask: “The batteries are new. Why won't it light?” Collect ideas on the board.
2. Teach (8 min): Draw the loop: battery, wire, bulb, and back to the battery. Explain voltage as the push and current as the flow. Show an LED's long and short legs, and why it needs a resistor.
3. Build (7 min): Each group tapes two AA batteries end to end, plus to minus, and connects the bulb (or the LED and its resistor) with wires, leaving a gap between two free ends. Touching the ends together should light it.
4. Test (15 min): Groups predict each object first, then touch both free ends to it and record “lights” or “stays dark” in a table. They copy their results into two columns on the board.
5. Wrap-up (5 min): Ask: why is a wire copper inside plastic? Exit ticket: one conductor, one insulator, and one safety rule.

**Before class, build and test one tester yourself. Use batteries only. Keep button batteries away from young children, and don't let pupils hold a bare wire across a battery: it gets hot.**

## Discussion

- Which result surprised you, and why?
- Why does an LED need a resistor when a torch bulb doesn't?
- Where could a short circuit happen by accident at school or at home?

## Exit ticket

- Name one conductor, one insulator, and one safety rule for circuits.

**Try it at school tomorrow**
Conductor hunt: Make a tester from two AA batteries, a torch bulb (or a red LED with a 100 Ω resistor) and three wires. Groups predict, then test ten everyday objects and sort them into conductors and insulators on the board. Finish by asking: why is a wire copper inside plastic?

Practice questions · for you

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**1** **Kemi wants her class's LED to be brighter, so she plans to take out its resistor. What should you tell her?**

- ◯ Go ahead: without the resistor it will be at its brightest
- ◯ Don't: it will burn out. Use a smaller resistor instead
- ◯ Don't: an LED without a resistor won't light at all

**2** **A torch with brand-new batteries and a working bulb must light up.**

◯ True ◯ False

**3** **Pupils touch each object across the gap in a battery-and-bulb tester. Which will light the bulb?**

☐ A coin ☐ A strip of kitchen foil ☐ A dry wooden stick ☐ A plastic ruler

☐ A rubber eraser ☐ A steel key

**4** **Choose a resistor for a red LED on a 5 V USB supply.**

A red LED takes 2 V, so _____ volts are left for the resistor. For 15 mA, the resistor needs about _____ ohms.

**Word bank:** 5 · 3 · 330 · 200

**5** **Match each part to its job in the circuit.**

| | | |
|---|---|---|
| Battery | • • • • | Turns current into light, one way round |
| Wire | • • • • | Limits the current |
| Resistor | • • • • | Gives the push (voltage) |
| LED | • • • • | Opens or closes the loop |
| Switch | • • • • | Carries current round the loop |

✎ **Which everyday objects from your pupils' homes could you use to teach circuits? Where could a short circuit happen by accident at school?**

Check your answers

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**1** **Don't: it will burn out. Use a smaller resistor instead**

Without a resistor nothing limits the current, so it shoots up and the LED burns out in a flash; it doesn't stay dark.

**2** **False**

Current flows only round a closed loop. A loose wire or a dirty contact breaks the loop, and a battery put in backwards cancels the push, so the torch stays dark however new its batteries are.

**3** **A steel key; A coin; A strip of kitchen foil**

The key, the coin and the foil are metals, so they conduct and close the loop.

**4** **A red LED takes 2 V, so 3 volts are left for the resistor. For 15 mA, the resistor needs about 200 ohms.**

5 − 2 = 3 V for the resistor, and 3 ÷ 0.015 = 200 Ω.

**5** **Battery — Gives the push (voltage)**
**Wire — Carries current round the loop**
**Resistor — Limits the current**
**LED — Turns current into light, one way round**
**Switch — Opens or closes the loop**

Every part has one job. When pupils can say each part's job, they can find the fault when a circuit doesn't work.